



NANYANG TECHNOLOGICAL UNIVERSITY

EE6222 Machine Vision

Assignment 1: Dimensionality Reduction for Classification

MSc in Electrical and Electronic Engineering

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1 Introduction

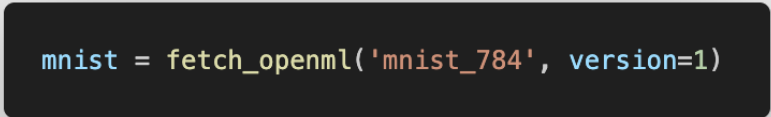
In this Assignment, I selected the MINIST dataset for training and testing the performance of classifiers. As for the dimension of images are large, I explored several dimensionality reduction methods to reduce the dimension of data while trying to retain the most important features, including **PCA**, **LDA**, combined method (Fisherfaces), **LLE**, **Isomap**, and **PaCMAP**. I compared the effectiveness of these methods by plotting their performance with varying number of dimensions.

In addition, I establish several basic classifiers, like **MMD**, **KNN**, and include Logistic Regression Classifier with sklearn package. Therefore, I not only compare the effectiveness of different dimensionality reduction methods, but also access the accuracy of classify images for these classifier methods. I would describe the details of my experiment in the following sections.

2 Dataset and Preprocessing

2.1 Dataset Description

The MINIST dataset contains **70,000** images of handwritten digits (0-9) with a resolution of 28x28 pixels, which means there are **10** classes intotal. Each image had been saved with the format of vector with **784 in length**. In addition, the images are only have one channel, which means they are grayscale pixels. Because of the pre-shaped vectors, single channel, and easy to get features, I think MINIST dataset is suitable for this assignment. Therefore, I fetch the dataset from sklearn package directly 1.



```
mnist = fetch_openml('mnist_784', version=1)
```

Figure 1: Fetch Code

2.2 Normalization and Standarlization

Since the MINIST dataset had been well preprocessed and tested by plenty of researchers, I didn't do too much basic preprocessing, like outlier and missing value detection. However, I though normalization and standarlization arer still neccessary for the following basic classifiers, which may be affected intensely by the scle of features. So I rearrange the vectors to 0 mean and unit variance.

$$\bar{X} = \frac{X}{L-1} \quad (1)$$

$$X_{std} = \frac{X - \mu}{\sigma} \quad (2)$$

2.3 Dataset Partition

I found the number of samples (70,000) is too large, which may lead to many difficulties in my following experiment. Firstly, I will training several classifiers with many dimension reduction methods, meaning the number of training process will be big. If I train 50,000-70,000 for each training, the consumed time will be very long. Secondly, my classifiers which will be used are simple and basic, therefore, they are easy to converge with small scale of data. With these two reason, I decide to extract 7,000 samples from the origin dataset formly to reduce the expence of time.

After that, I split the dataset into training set and testing set through *train_test_split* function with ratio of 6:1 and finished the process of preparation of dataset 2.

```
Training Feature Dimension: (6000, 784)
Traing Label Dimension: (6000,)
Testing Feature Dimension: (1000, 784)
Testing Label Dimension: (1000,)
```

Figure 2: Dimension of practical dataset

3 Methodology

3.1 Dimensionality Reduction Methods

3.1.1 Linear Discriminant Analysis

Linear Discriminant Analysis (LDA) aims to find a linear combination of features with weight matrix $\mathbf{w}$ to transform the data into linear space, ensuring that the separation between different classes is maximized while the variance within each class is minimized.

Step 1: Mean Value Computation

With given m classes $\{w_1, w_2, \dots, w_m\}$, there are c samples in each class w : $\{X_1, X_2, \dots, X_c\}$. For each sample vector, it has n features, $\{x_1, x_2, \dots, x_n\}$. In this experiment, $n=784$.

Overall Mean:

$$\mu_t = \frac{1}{N} \sum_{i=1}^N x_i \quad (3)$$

Class Mean:

$$\mu_i = \frac{1}{n_i} \sum_{j=1}^{n_i} x_j \quad (4)$$

Where N is the total number of samples, and n_i is the number of samples in class w_i .

Step 2: Scatter Matrix Computation

Between-Class Scatter Matrix:

$$S_B = n_i * \sum_{j=1}^m (\mu_j - \mu_t)(\mu_j - \mu_t)^T \quad (5)$$

Within-Class Scatter Matrix:

$$S_W = \sum_{j=1}^m \sum_{x_i \in w_j} (x_i - \mu_j)(x_i - \mu_j)^T \quad (6)$$

```
Sw = np.zeros((n_features, n_features))
for i, c in enumerate(self.classes):
    Xc = X[y == c] - means[i]
    Sw += Xc.T @ Xc
Sw += 1e-6 * np.eye(n_features) #invertable
Sb = np.zeros((n_features, n_features))
for i, c in enumerate(self.classes):
    n_c = (y == c).sum()
    mean_diff = means[i] - self.overall_mean.reshape(-1, 1)
    Sb += n_c * (mean_diff @ mean_diff.T)
```

Figure 3: Scatter Computation Code

Step 3: Weight Matrix Computation

As I mentioned before, the LDA aims to find the optimal weight matrix to separate classes on two dimension, equaling to the problem: maximize the ratio of between-class scatter to within-class scatter.

$$\mathbf{w}^* = \arg \max_{\mathbf{w}} \frac{|\mathbf{w}^T S_B \mathbf{w}|}{|\mathbf{w}^T S_W \mathbf{w}|} \quad (7)$$

Since the $S_B \mathbf{w}_i = \lambda_i S_W \mathbf{w}_i$,

$$\lambda \mathbf{w} = S_W^{-1} S_B \mathbf{w} \quad (8)$$

Therefore, the $\mathbf{w}$ is the eigenvectors matrix according to eigenvalue λ , the largest λ refers to best matrix $\mathbf{w}^*$.

```
eigvals, eigvecs = eigh(Sb, Sw)
sorted_idx = np.argsort(eigvals)[::-1]
max_comp = min(len(self.classes) - 1, n_features)
n_comp = max_comp if self.n_components is None else min(self.
n_components, max_comp)
self.scalings = eigvecs[:, sorted_idx[:n_comp]]
```

Figure 4: Eigenvector Matrix Computation

Step 4: Transform Dataset

After getting the weight matrix, the transformation of input data $\mathbf{X}$ is:

$$\mathbf{Y} = \mathbf{X}\mathbf{w}^* \quad (9)$$

In practice, we will choose the k^{th} value of sorted eigenvalues to form $\mathbf{w}^*$. Theoretically, LDA is designed to project samples onto a lower-dimensional subspace, with the dimensionality being at most $C - 1$, because the rank of S_B is at most $C-1$, C is the number of classes.

3.1.2 Principal Component Analysis

Similar to LDA, Principal Component Analysis (PCA) is another effective method to dimensionality reduction. The difference is PCA can project the input data to different dimensions without limitation like LDA. The projected data after PCA will also separate largely, which means the redundancy elements are eliminated.

Step 1: Data Processing

Before the projection, the data should be scaled and normalized, eliminating bias. Specifically, compute mean value to shift to 0 mean and covariance to investigate the distribution situation of data.

Covariance Matrix Computation:

$$\Sigma_i = \sum_{x_j \in w_i}^{n_i} (x_j - \mu_i)(x_j - \mu_i)^T \quad (10)$$

Where μ_i is the mean value of class w_i

Step 2: Eigenvectors Computation

The PCA aims to find the direction $\mathbf{w}$, the variance of data projected to that direction is largest. For the transformed data $Y = \mathbf{w}X$, variance computed:

$$Var(Y) = \frac{1}{N} \sum_{i=1}^N (w^T x_i)^2 = w^T \Sigma w \quad (11)$$

Solving the $\arg \max_w (w^T \Sigma w)$,

$$\Sigma \mathbf{w} = \lambda \mathbf{w} \quad (12)$$

Larger eigenvalue refers to better eigenvector, by sort the eigenvalues in descending sequence, we can fetch k^{th} large vlues and their correspondingly eigenvectors to form best matrix $\mathbf{w}^*$.

```

self.mean = np.mean(X, axis=0)
X_centered = X - self.mean
covariance_matrix = np.cov(X_centered, rowvar=False)
eigenvalues, eigenvectors = np.linalg.eigh(covariance_matrix)
sorted_indices = np.argsort(eigenvalues)[::-1]
self.components = eigenvectors[:, sorted_indices[:self.
n_components]]

```

Figure 5: PCA Realization Code

Step 3: Transform Dataset

With above process, we can get the corresponding optimal weight matrix, according to the input data and component coefficient k . Applying the matrix to input data, transformation finished.

3.1.3 Fisherfaces

Based on learning in course, I acknowledge that the reduction of feature equals to deleting redundant features. As for LDA, it can only remain $C-1$ features, which causes the low accuracy after LDA. Whereas, the PCA with large component m

3.1.4 Locally Linear Embedding

3.1.5 Isometric Mapping

3.1.6 Pairwise Controlled Manifold Approximation

3.2 Classifiers Establishment

3.2.1 Minimum Mahalanobis Distance Classifier

3.2.2 K-Nearest Neighbors Classifier

3.2.3 Logistic Regression Classifier

3.3 Evaluation Methods

4 Results and Discussion

5 Conclusion

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References

- [1] I. K. Fodor, “A survey of dimension reduction techniques,” Tech. Rep. UCRL-ID-148494, Lawrence Livermore National Laboratory, Livermore, CA, USA, May 2002.

Appendix